

Research paper

Machine learning-based uncertainty analysis in power system planning: Insights and pathways for decarbonization

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ABSTRACT

This study performs an extensive uncertainty analysis of decarbonization pathways for the Canadian power system. By leveraging machine learning (ML) techniques, we represent complexities and nonlinear behaviors of the power system, which are challenging to observe using traditional modeling methods due to computational limitations. We further utilize K-means clustering and statistical methods to identify correlations, investigate the effects of uncertainty in key parameters (capital costs, demand, and carbon tax), and identify opportunities for diversification of the power generation mix. Key findings include the increased flexibility of gas combined cycle (generation mean 4.8 TWh, installed capacity mean 4.8 GW) over gas simple cycle (generation mean 22.3 GWh, 0 installed capacity) in response to carbon tax, highlighting its role in a low-carbon future. Variations in solar and wind deployment rates, influenced by capacity factors and location-specific costs, underscore the necessity of a flexible grid system. This analysis reveals important correlations between renewable deployment, and transmission expansion and gas combined cycle reliance. The results also highlight minimal sensitivity of hydro-electric power generation (~43 % of scenarios) to changes in input parameters. The novelty of our work lies in proposing a comprehensive approach which uses ML techniques to go beyond the capabilities of existing models, specifically by performing extensive uncertainty analysis and provides key insights into system behavior, causative factors, and an expanded breadth of potential pathways. Taken together, this proposed approach lays a foundation for strategic planning and policy formulation.

1. Introduction and background

The decarbonization of power grids represents a monumental challenge in the field of energy systems due to its inherent complexity. These advanced systems require the integration of various elements, including renewable energy resources, advanced metering infrastructure, energy storage, and sophisticated real-time data analytics, all of which need to operate in sync to ensure system reliability, efficiency, and resilience. This complexity is amplified by the challenge of transitioning power systems towards renewable resources, which brings with it greater inherent variability among other sources of uncertainty (Farhangi, 2010).

Understanding these complexities, as outlined by Giampietro et al ((Mario) Giampietro et al., 2013), is critical for effective policymaking, increasing both resilience and reliability. Identifying leverage points, or human-controllable parameters that can significantly impact system

behavior (Leverage Points, 2023), is essential for guiding energy system transformations.

1.1. Modeling process diagram

Quantitative models serve as simplified representations of real-world systems, necessarily focusing on selected aspects while disregarding others. Therefore, models have inherent prediction error and are only meaningful within boundaries, defined by key variables and relationships, and specific boundary conditions, creating a "finite descriptive domain" (Motesharrei et al., 2016; Grubler et al., 1999; Harrison, 2010). The arbitrary process of creating a model, subject to modeler bias and methodological limitations, produces varying degrees of abstraction relative to the actual system. This can be increased further by methods used to improve computational tractability. For instance, surrogate modeling, which focuses on isolating crucial parameters and replicating

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relationships, involves both additional abstraction and error. This trade-off between computational efficiency and modeling precision requires careful model evaluation, selection, and use, informed by the relevant analytical goals. Although models have inherent shortcomings and cannot perfectly reproduce real-world behavior, they nevertheless can be used to provide valuable insights into complex systems. As statistician George Box (Smith, 2015) famously said, "all models are wrong, but some are useful."

These considerations are particularly relevant when researching complex socio-technical systems, whose stochastic nature emphasizes the need for uncertainty analysis. Models are subject to both aleatory and epistemic sources of uncertainty. As such, analyzing model results requires explicit and adequate treatment of these uncertainties. Increasingly, uncertainty and sensitivity analysis emerging as key components of uncertainty management in modeling best practice. Monte Carlo simulation, ensemble modeling, and related methods, commonly used in fields such as meteorology, offer comprehensive approaches to uncertainty analysis. Notably, the study of future energy transitions is in its early stages ((Mario) Giampietro et al., 2013; Loftus et al., 2015), uncertainty analysis is set to play a vital role in providing realistic assessments of future scenarios, acting as a bridge between complex system modeling and effective planning.

Uncertainty analysis methods primarily focus on evaluating the variability of model outputs to characterize potential behaviors of the system. This process involves quantifying the uncertainty in model outputs stemming from input parameters and modelling assumptions, among other sources, aiming to gain a better understanding of the plausible ranges and probability distributions of modelled outcomes. Additionally, uncertainty analysis is a prerequisite to sensitivity analysis, which builds on uncertainty analysis to map specific causal influences between inputs and output uncertainties (Saltelli et al., 2019; Refsgaard et al., 2006). Fig. 1 provides a visual representation of the relations between real-world systems, models, and approaches to uncertainty discussed in this section.

Comprehensive uncertainty analysis in energy systems modeling is crucial for addressing emerging complex, multi-faceted problems exhibiting interconnected social, economic, and environmental relationships, such as decarbonization of the energy system. Uncertainty analysis can help to navigate this complexity by providing an improved understanding of plausible outcomes, non-intuitive system behaviors, and trade-offs between desirable outcomes.

The importance of climate action and the associated potential for unintended consequences further emphasize the importance of uncertainty analysis in energy systems analysis, including electric power

systems. Unforeseen outcomes often arise from system interdependencies, challenging initial expectations. For example, the increased use of renewable energy resources may not lead immediately to a decrease in greenhouse gas emissions and could also lead to the intensification of land use and stretch the availability of critical materials. Such potentialities cannot always be anticipated using selected quantitative representations, implying greater pluralism in energy systems modeling and analysis is crucial. Uncertainty analysis is one method to enhance the exploration of possible outcomes, supporting diverse perspectives in energy systems research and decision-making (Refsgaard et al., 2006; Bayer, 2023; Walker et al., 2013).

1.2. Uncertainty analysis for power systems

Uncertainty and sensitivity analysis are increasingly recognized as crucial components for the quantitative modelling of complex socio-technical systems. This trend has prompted suggestions for standardizing their application in various fields (Saltelli et al., 2019; Refsgaard et al., 2006). Uncertainty analysis provides substantial value in informing decision-making and policy formulation, while also broadens the conceptual space of possibilities, improves the transparency of quantitative modelling efforts, and acts as an essential tool for the promotion of modeler reflexivity (Saltelli et al., 2019). Power systems exhibit pronounced complexity due to their long-time horizons, significant cumulative capital investments, interlinked social, technical, and ecological aspects, and their centrality to economic development. Uncertainty analysis can generate new insights for planning and enhancing the flexibility and resilience of power systems, as it aids in quantifying the level of confidence attributable to a given result within a broader range of plausible outcomes.

While uncertainty analysis has not been extensively applied in energy systems modelling, its importance is highlighted by the growing body of literature on the topic. For example, Gabrielli et al (Gabrielli et al., 2019). presented a framework for designing multi-energy systems with limited data. Their application to a Swiss urban neighborhood emphasized the criticality of weather condition uncertainties for optimal design prioritizing cost and emission reduction. Mavromatidis et al (Mavromatidis et al., 2018). investigated uncertainty in distributed energy systems design. Their Swiss case study revealed notable variations in the economic and carbon performance of optimal Distributed Energy Systems designs. Energy demand patterns and carrier prices emerged as significant drivers of cost variability. Aien et al (Aien et al., 2016). provided a comprehensive review of uncertainty modelling for electric power system studies, highlighting the strengths, weaknesses, and

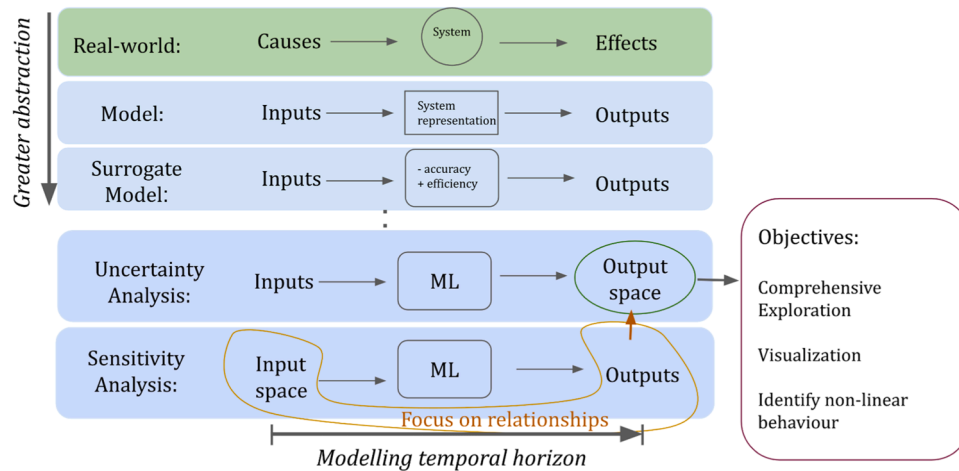


Fig. 1. Real-world systems must be abstracted to produce models, introducing uncertainty, particularly where techniques are required to improve computational efficiency. While efficient models are well-suited for comprehensive uncertainty analysis, and subsequently sensitivity analysis, the associated uncertainty must be explicitly managed.

applicability of different methods. Soroudi and Amraee (Soroudi and Amraee, 2019) proposed a standard classification of uncertainty modelling techniques in energy system studies, evaluated their strengths and weaknesses, and suggested areas for future uncertainty analysis research. Yue et al (Yue et al., 2018). assessed uncertainty in energy system optimization models, focusing on four uncertainty approaches and their effectiveness in enhancing the policy relevance of model results. These examples demonstrate the growing relevance and importance of uncertainty analysis in energy systems modelling, despite clear gaps remaining, particularly regarding established uncertainty analysis standardization and utilization.

Uncertainty analysis in energy systems modeling, while offering valuable insights, poses computational challenges. The most prominent challenge, commonly known as the 'curse of dimensionality,' arises due to the extensive computational cost generated by the multifaceted nature of uncertainties. Added to the complexity are non-linear relationships between input parameters and model results, real-world uncertainty incorporation into modeling, and the estimation of rare events and extreme outcomes. These aspects highlight the necessity for strategies to minimize computational costs.

The complexity of power systems amplifies the computational cost of detailed models, particularly when global optimization strategies are employed, as in the planning models like COPPER (Arjmand and McPherson, 2022a). This condition emphasizes the need for the development of high-quality surrogate modelling techniques such as the one developed by Jahangiri et.al (Jahangiri et al., 2023a) to reduce computational requirements without significantly compromising accuracy.

The inherent complexity and heavy computational demands of power systems modeling make the traditional methods of uncertainty analysis inadequate and call for the use of advanced, efficient analytical methods, such as Machine Learning (ML). ML algorithms can be trained on large data sets, enabling them to adapt and improve over time. This allows for more precise replication of their underlying models. ML also provides solutions to the non-linear aspects of power systems. With their skill for discovering complex patterns and dependencies, ML algorithms deliver detailed representations of the system's behavior.

ML contributes significantly to the development of surrogate models. These models, serving as simplified versions, can capture the complex relationships and behaviors within energy systems while using less computational power. Surrogate models developed with ML enable thorough and efficient uncertainty analysis and sensitivity analysis. Capable of navigating high-dimensional input spaces, these models extensively examine the variability and sensitivity of model results, providing a strong understanding of system behavior across a range of scenarios and input uncertainties. Moreover, ML supports the systematic exploration of the output space, efficiently identifying key areas of interest and significant uncertainties.

ML has also been widely used for uncertainty analysis in energy systems. Gaussian process regression models have been applied to develop surrogate models of energy systems. Najibi (Najibi et al., 2021) used gaussian process regression for probabilistic short-term solar photovoltaic output forecasting and decreased error by 55.5 % compared to alternative methods. Bayesian methods are another ML technique for uncertainty analysis. Zhu et al (Zhu et al., 2020). proposed a new Bayesian method to calibrate energy models of buildings. Sun et al (Sun et al., 2020). used the Bayesian method to predict net load subject to stochasticity arising from distributed photovoltaic generation in Australia. ML models have also been deployed to manage uncertainties in energy management and planning in power systems. Aguilar et al (Aguilar et al., 2021). proposed an optimization method for distributed energy resource management using ML techniques to handle uncertainty related to generation and load profiles. Results showed that the proposed scheme mostly works better than deterministic approaches in terms of penalty reduction, demonstrating its effectiveness in managing uncertainty for distributed energy resources (C. C. Energy Regulator,

2021) Finally, Razavi et al (Razavi et al., 2021). outlined areas warranting further attention for systems modeling, including standardizing sensitivity analysis as a discipline, addressing the computational burden, progressing sensitivity analysis in the context of ML, and clarifying the relationship between sensitivity analysis and uncertainty quantification.

In summary, the existing literature highlights the critical role of uncertainty analysis in identifying interdependencies among energy system variables, especially in complex, large-scale power systems. These interdependencies encompass factors such as generation capacity, transmission lines, load demand, and renewable energy penetration. However, traditional uncertainty analysis methods, being computationally demanding and time-consuming, limit their applicability to decision-makers needing prompt and accurate information. Gaps persist in exploring the application of ML to analyze complex power systems design space, which includes numerous input parameters and output variables. While current ML applications primarily focus on optimization and prediction, there is a lack of research into the exploratory use of ML and investigations into output space correlations, which could offer valuable insights into power system behavior. This work closes that gap by utilizing Deep Neural Network (DNN) (Jahangiri et al., 2023a) as surrogate models for conventional, high-cost simulation models, which is crucial for enabling more robust uncertainty analysis in large-scale power systems. This approach promises a comprehensive understanding to inform effective policy planning and decision-making. Nonetheless, careful consideration must be given to specific analytical objectives and associated techniques, given the highly nonlinear characteristics of real-world power systems. A critical step involves a thorough exploration of the output space of power system models, encompassing all plausible outcomes represented by significant output variables at discrete time steps. Thus, this paper aims to contribute to standardizing uncertainty analysis in complex socio-technical systems by extending these efforts to the power systems domain using ML methods.

1.3. Contributions

Existing literature emphasizes the importance of uncertainty analysis in understanding energy systems, particularly in complex power systems. However, traditional methods are computationally intensive, requiring more efficient approaches to support decision-making. The gap that currently exists in the literature is the limited use of ML for uncertainty analysis, particularly exploring complex power system design spaces with numerous input and output variables. Most ML applications focus on optimization and prediction, lacking research into exploratory use and output space correlations.

This work fills that gap by introducing a novel blend of uncertainty analysis and ML in large-scale power system planning for decarbonization, addressing the notable gap in understanding the intricate interconnections and non-linearities in system behavior over time. The originality of this paper is that it uses ML to circumvent computational limitations to characterize model behavior and explores both input and output spaces comprehensively to identify key correlations and non-linearities in system behavior, providing a more nuanced understanding essential for effective policy planning. The contributions of this work are as follows:

- **Leveraging ML for Comprehensive System Modeling:** characterizing the full scope of model behavior using ML to circumvent computational limitations.
- **Input Domain Analysis:** identifying the range, variability and uncertainty of key model inputs; including demand, capital costs, and carbon tax.
- **Output Space Variability:** identifying the range and variability of key model outputs given the uncertainty in model inputs, allowing for the variation of input parameters along multiple dimensions simultaneously.

- **Enhanced Uncertainty Analysis:** presenting and analyzing model results, plausible ranges and probability distributions to fully characterize system behavior and discuss the implications.
- **Analyzing System Behavior:** uncovering and characterizing nonlinearities in the output space, regarding critical thresholds, system bifurcation, etc.

By embracing these advanced tools and methods, we aim to take the first step in revolutionizing energy systems planning, transitioning from conventional approaches to more capable, modern techniques that drive impactful advancements in the field.

2. Methodology

In this study, a previously developed deep learning method is utilized to investigate the extensive design possibilities of the Canadian power system. The method, (U. U. Energy Information Administration, 2020), is employed to construct a surrogate model based on the COPPER model (Jahangiri et al., 2023b) for uncertainty analysis of the Canadian power system. The limitations of the surrogate model are addressed by expanding the range and number of input parameters, enabling a more comprehensive examination of uncertainty related to model outcomes. Additionally, we overcome the constraints, storage limitations, and national output issues of COPPER V5 by incorporating COPPER V7, along with the inclusion of transmission expansion.

Once the DNN is trained, it serves as a surrogate model for COPPER in uncertainty analysis. Specifically, by providing a set of identified uncertainties in the input variables, the DNN can rapidly and accurately predict the COPPER simulation representing the projected state of the system. This approach reduces computational cost of COPPER, enabling us to explore broader pathways and conduct multidimensional sensitivity and uncertainty analyses. By leveraging surrogate model, we can uncover non-intuitive correlations within the design space and conduct comprehensive scenario exploration.

The ML-based surrogate model employed a dataset of COPPER simulations for training process. We identified critical uncertain inputs, including the carbon tax, capital costs for different generation types, and annual demand growth rate by province, which exhibit potential for variability, thereby exerting significant influence on model outputs. These parameters serve as boundary conditions defining potential trajectories within the Canadian power sector.

To capture a comprehensive range of uncertainties related to capital costs, we expanded the existing capital cost ranges based on insights from relevant literature. We used data from six distinct sources and transformed the estimates provided by each source into equivalent annual payments over a predefined amortization period, denoted as CAD\$/kW/year, in line with the original COPPER model formulation (Arjmand and McPherson, 2022a; Jahangiri et al., 2023b; C. C. Energy Regulator, 2021; U. U. Energy Information Administration, 2020, 2022; Projected Costs of Generating Electricity, 2020; Utility-Scale Battery Storage, 2023; Electricity, 2023). Next, we calculated the interquartile range (IQR) using the mean and standard deviation across all these estimates. By considering the extremes of the identified IQR and the initial range limits, as outlined in (Arjmand and McPherson, 2022a), we updated the range limits accordingly. In the case of battery storage, given the presence of various chemistry types, we accounted for the capital cost associated with several chemistries, including Li-Ion (LFP), Li-Ion (NCA), Li-Ion (NMC), Vanadium flow, and ZnBr.

As per the existing policy framework (How carbon pricing works - Canada.ca, 2023), the current carbon pricing legislation mandates an annual price increase of \$15 per ton, starting from \$65 per ton in 2023 and reaching \$170 per ton by 2030, with no regulation beyond this date. Under the lower bound scenario, prices rise to \$95 per ton by 2025 (the base year for COPPER's carbon pricing) and then remain constant. The upper bound scenario projects prices to increase to \$95 per ton by 2025, then accelerate by \$30 per ton annually to \$245 per ton by 2030, and

subsequently increase by \$20 per ton each year, reaching \$645 per ton by 2050. This modification allows the examination of the effects of both higher and lower carbon taxes on various aspects of the power system, including emissions, generation capacities, and transmission expansion.

Demand growth varies between provinces and was taken from Canada's Energy Future Report (Canada Energy Regulator, 2023a) for 2050, with 2018 as the base year, and it was also expanded by 25 % on either side of the range, which is a common practice for sensitivity analysis. This approach allows for a wider range of potential demand outcomes, capturing the significant variability and unpredictability associated with demand fluctuations (Canada's Energy Future, 2021). The final ranges for these parameters can be found in Table 1. To ensure a comprehensive exploration of the input space, we utilized Latin-hypercube sampling (LHS) to sample these parameters across the specified ranges outlined in Table 1.

The COPPER model was run using these inputs, without imposing additional restrictions: constraints are not placed on national emission limit and the development of transmission, which was limited to a maximum increase of 900 %. This approach allowed the model to expand transmission within the bounds of current land use and policy constraints while minimizing limitations. Consequently, a comprehensive range of feasible scenarios was generated. The focus of this study is on the key national outputs for the year 2050, including carbon emissions, the types and capacities of installed power generation, the supply from each power generation source, and the installation of interprovincial transmission lines. Examining these selected outputs, provide a holistic view of the power system's behavior.

Following data generation, the data underwent manipulation, including data cleaning, normalization, and variance clipping. Through the empirical model development procedure, the optimal hyper-parameters for the model architecture were selected (Jahangiri et al., 2023b).

2.1. Uncertainty analysis using surrogate model

To investigate the behavior of the model outputs, all input parameters are simultaneously populated using LHS for 2000 sample points, as further sampling did not yield any discernible changes in behavior. Then, these samples are supplied to the surrogate model to generate corresponding outputs, effectively promoting a comprehensive exploration of the output space.

2.1.1. Distribution histograms

Histograms of output parameters represents how they spread by observing the shape, center, and span of the distribution, which can infer about the system's behavior under different conditions. These histograms visualize the characteristics and underlying probability distribution of each output variable, revealing information about normality, skewness, and multimodality. This information is crucial as it forms the basis for decision-making, risk analysis, and further statistical analysis.

Table 1

Selected ranges for input variables based on 6 references to include uncertainties.

Input variable	Range	Unit
Carbon tax in 2050	95 – 645	\$/ton
Total demand growth 2050/2018	1.2 – 2.5	Ratio
Annualized capital cost of gas combined cycle (gas CC)	106 – 198	\$/kW
Annualized capital cost of diesel	104 – 195	\$/kW
Annualized capital cost of coal	289 – 549	\$/kW
Annualized capital cost of nuclear	536 – 972	\$/kW
Annualized capital cost of biomass (waste)	285 – 574	\$/kW
Annualized capital cost of gas simple cycle (gas SC)	84 – 145	\$/kW
Annualized capital cost of wind	84 – 242	\$/kW
Annualized capital cost of solar PV	86 – 216	\$/kW
Annualized capital cost of battery storage	60 – 116	\$/kW

The generated distribution histograms enhance our understanding of the real-world system's potential behavior, capturing the variability and uncertainty in model outputs when considering input uncertainties. By exploring a wide range of input scenarios, we gain valuable insights into the system's dynamics and the range of possible outcomes.

Measures of central tendency, such as the mean or median, provide estimates of the most probable or average output values. The spread of the distribution, indicated by the standard deviation (std) or IQR, reflects the level of variability or uncertainty in the output. Skewness measures the distribution's asymmetry, with positive skewness indicating a longer right tail and negative skewness indicating a longer left tail. Outliers or extreme values highlight potential anomalies or exceptional cases.

A concentrated distribution signifies high certainty and reliable predictions. A multimodal distribution indicates that there are several popular or common values (or ranges of values) for this parameter across different scenarios, showing system bifurcations, leading to mutually exclusive outcomes with significant policy implications. It is crucial to interpret output distributions within the context of the specific problem and analysis goals.

2.1.2. Quantifying relationships in the output space

Investigating correlations between output variables allows us to identify significant relationships and quantify system tendencies as power systems evolve over time. These correlations provide a systems-level perspective on potential outcomes, ranging from expected and obvious to non-intuitive patterns that reveal underlying dynamics often overlooked in energy transition discourse.

To comprehensively investigate output space interdependencies and correlations, we employ two metrics. One of these metrics is the Mutual Information (MI) score, which measures the statistical dependence between two random variables. Unlike correlation, MI captures any type of statistical dependency, including non-linear and non-monotonic relationships. It quantifies the amount of information gained about one variable by observing the other, effectively reducing uncertainty. The MI score ranges from zero (indicating independence) to larger values (signifying greater dependency). The shared information between two variables, X and Y , is defined as Eq. (1) (Janssen et al., 2023).

$$MI(X; Y) = \sum_X \sum_Y P(X, Y) \log \left(\frac{P(X, Y)}{P(X)P(Y)} \right) \quad (1)$$

$P(X, Y)$ is the joint probability mass function of X and Y . $P(X)$ and $P(Y)$ are the marginal probability mass functions of X and Y respectively.

Another metric used is Spearman's rank correlation coefficient, commonly referred to as Spearman's rho (ρ). This statistical measure assesses the strength and direction of the non-linear monotonic relationship between two variables. For a sample of size n , the n raw scores X_i , Y_i are converted to ranks $R(X_i)$, $R(Y_i)$, and r_s is computed as Eq. (2) (Stephanou and Varughese, 2021).

$$r_s = \rho_{R(X), R(Y)} = \frac{\text{cov}(R(X), R(Y))}{\sigma_{R(X)} \sigma_{R(Y)}} \quad (2)$$

ρ denotes the Pearson correlation coefficient, but applied to the rank variables, $\text{cov}(R(X), R(Y))$ is the covariance of the rank variables, $\sigma_{R(X)}$ and $\sigma_{R(Y)}$ are the standard deviations of the rank variables.

These metrics have their own strengths and limitations, necessitating the consideration of multiple factors for a comprehensive exploration. While Spearman's rank correlation coefficient provides information on the direction of the correlation, it cannot detect non-monotonic relationships, unlike the MI score. The combination of these metrics provides a robust means to identify associations between output variables, making minimal assumptions regarding their form.

2.1.3. Feature importance clustering

The output space comprises national outputs, which serve as the

features for each of the 2000 samples in our dataset acquired from the surrogate model. To gain further insights into this dataset, we employed a technique called K-means feature importance clustering, which groups samples based on their feature values (Liu et al., 2010). By utilizing the K-means algorithm, we identify patterns and similarities among the samples in our dataset. This approach assigns an importance factor to each output within each cluster. By extracting the most significant output variables across all groups, we understand how these variables contribute to the overall behavior of the system. This information is valuable for comprehending system behavior under different scenarios.

3. Results

3.1. Model development

During model training, the first metric we monitor is the Mean Squared Error (MSE) loss, which indicates the accuracy of the predictions. A lower MSE loss generally suggests greater accuracy in the model's predictions. However, it is not always clear what MSE loss value should be considered a sufficiently close approximation of the base COPPER model. To resolve this, we also assess the quality of the predictions by comparing the surrogate model outputs with the actual results from the COPPER model using the R-squared value as a secondary metric. The R-squared value measures the goodness of fit, with 1 indicating a perfect fit and 0 indicating no fit. For each datum in the test dataset, we calculate the R-squared value between the COPPER output and the surrogate model output, then derive the minimum, maximum, and mean R-squared values across the test set. Our acceptance threshold for model performance is set at a minimum R-squared value of 0.94. The performance results, including training and test errors, are summarized in Table 2. These results indicate a high degree of accuracy in the model's predictions (Jahangiri et al., 2023a).

Table 3 compares the proposed model's run-time and resource use with COPPER, showing significantly reduced computational costs and resource requirements. This enhanced efficiency enables more feasible scenario exploration without specific computational constraints.

3.2. Uncertainty analysis

Histograms of output distributions provide high-level insights on the outputs behavior by presenting their range and tendencies. The output statistics in Fig. 2 provide a snapshot of future scenarios for the power system, supporting effective planning for sustainable power infrastructure. The distribution plots of different output variables can be categorized as multimodal, spread with central or extremal tendency, unimodal, and largely non-stochastic.

Fig. 3 and Fig. 4 exhibit the MI score and Spearman's rank values respectively for variables within the Canadian power system (the non-stochastic variables in the histograms are removed from the exploration of interdependence).

The key outputs in shaping scenario differences and ultimately leading to the diverse pathways are explored through the feature importance clustering.

The silhouette score (Selecting the number of clusters with silhouette analysis on KMeans clustering, 2023) is used as a metric to identify the optimal number of clusters. A higher silhouette score, closer to one, indicates a more cohesive clustering arrangement. Accordingly, we determine that 16 clusters align with the maximum silhouette score. Fig. 5 depicts a bar chart of the importance values for each output in

Table 2
Training and Test MSE losses of the model and its accuracy in predictions.

Training loss (MSE)	Test loss (MSE)	Mean (R^2)	Std (R^2)
0.0039	0.01	0.96	0.020

Table 3
Comparison of COPPER and the proposed model computational costs and run time.

Model	CPU	Memory	Run-time
COPPER	16–32	64–128 GB	15 h – 72 h
Residual neural network	1	1.5 MB	10 ms

clustering. Higher weights indicate higher deterministic outputs in clustering and consequently shaping different pathways.

In the following section, we present a selection of insights derived from analytical approaches, including histogram distributions, correlation heatmaps, and clustering results.

3.3. Technologies with robust evolution

The evolution of several technology types remains robust against a breadth of input assumptions. Certain variables exhibited distributions with considerable spread around central or extreme values, suggesting a wide range of plausible output. This kind of distribution implies that there is a fairly robust pathway for a specific generation technology. Varying inputs does change the trajectory of that technology but around the central pathway. The central tendency indicates the "typical" or most likely level of that output. Extremal tendencies, on the other hand, represent the potential maximum and minimum output. These variables include national carbon emission, total cost, new installed capacity, and generation of solar and wind, new installed transmission of Prince Edward Island-New Brunswick, and new installed capacity and generation of gas combined cycle (gas CC). We delve into detail for some of

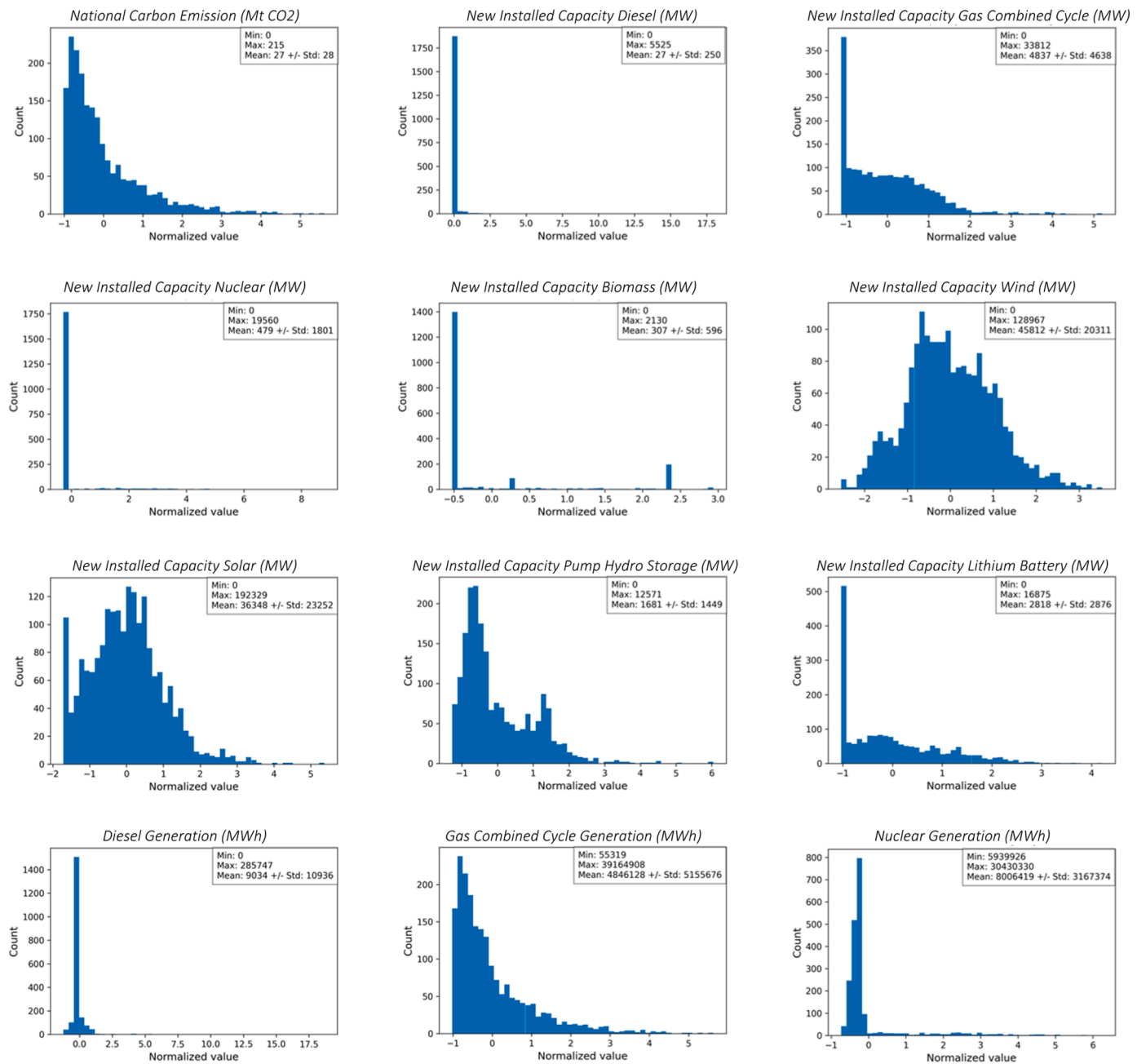


Fig. 2. Distribution of projected output variables of the Canadian power system in 2050 (Negative bins do not mean negative values of variables, they show values lower than average).

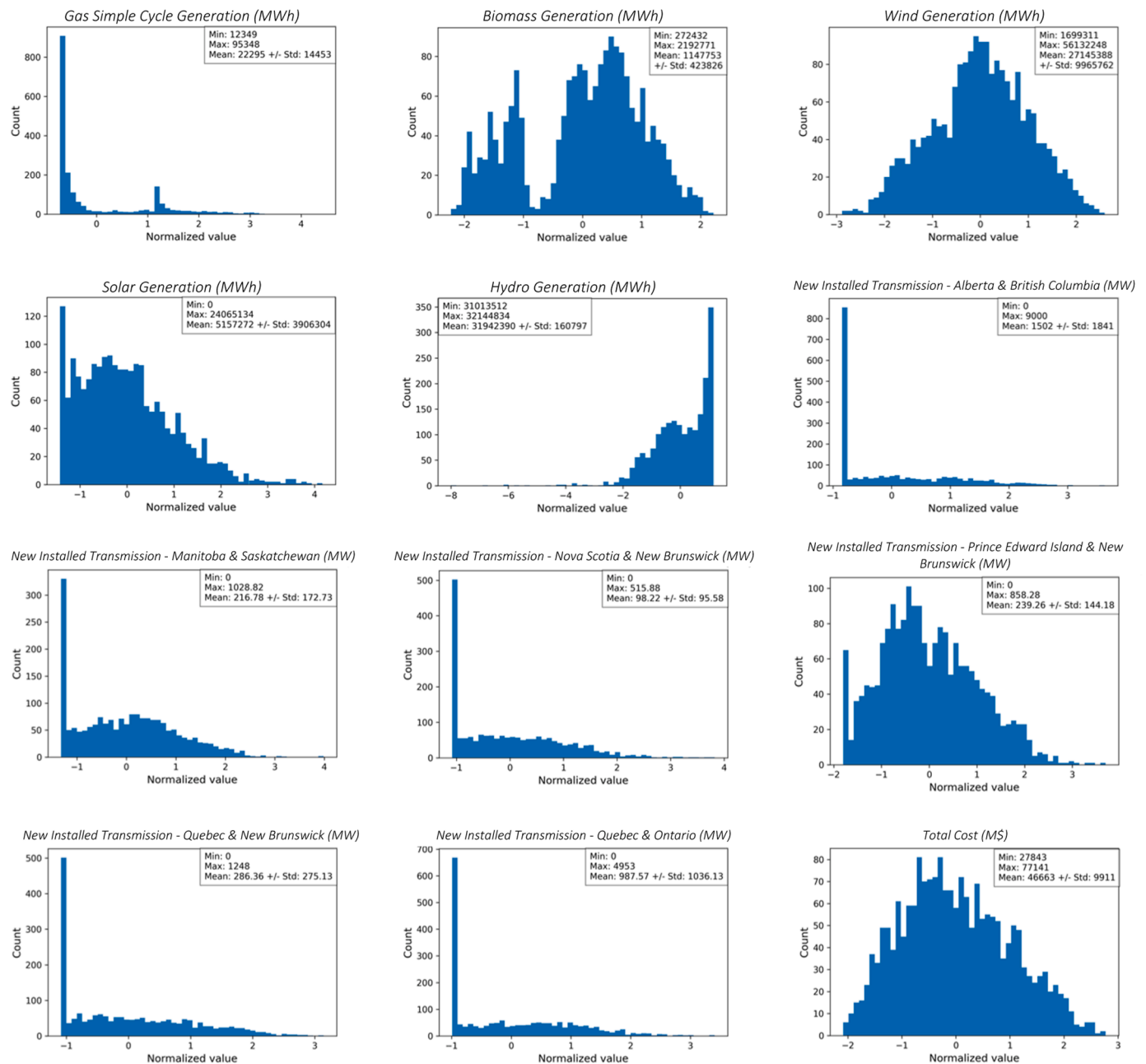


Fig. 2. (continued).

the more significant points:

A spread in the distribution suggests variability in the gas CC's generation (min 55.3 GWh, Max 39.1 TWh, mean 4.8 TWh, std 5.1 TWh) and new installed capacity (min 0, Max 33.8 GW, mean 4.8 GW, std 4.6 GW). The extremal tendency represents potential maximum or minimum outputs, reflecting the plant's capacity to ramp up during peak demand or scale down when other resources are plentiful. Gas CC positively correlates with both solar and wind, implying that these generation types might be used in tandem to balance the intermittent nature of renewable energy.

A wide distribution with both central and extremal tendencies for solar generation (min 0, Max 24 TWh, mean 5.1 TWh, std 3.9 TWh) and new installations (min 0, Max 192 GW, mean 36 GW, std 23 GW) indicates a wide range of plausible scenarios, from moderate to extreme. The central tendency suggests that there are scenarios where solar generation and capacity installations are close to the average, implying typical or expected conditions. Meanwhile, the extremal tendency

indicates that there are also situations where solar generation and new installations could be significantly lower or higher than the average, signaling possible extreme events, but they are rare and only happen in the limited combination of inputs. There aren't many instances in which solar doesn't follow the central trend, therefore there is a robust pathway.

The role of solar in differentiating various scenarios is discovered to be more significant than that of wind. This observation aligns with the unique distribution patterns of solar outputs, unlike wind which exhibits only central tendencies. The presence of these dual tendencies in solar indicates its significant influence on decision-making and strategy formulation.

A wider distribution of wind compared to solar implies that wind generation (min 1.6 TWh, Max 56.1 TWh, mean 27.1 TWh, std 9.9 TWh) and new installation (min 0, Max 128.9 GW, mean 45.8 GW, std 20 GW) have even more variability. The high central tendencies show the most likely outcomes (~ 68 %) are concentrated around the average scenario and implies a significant role for wind power in the

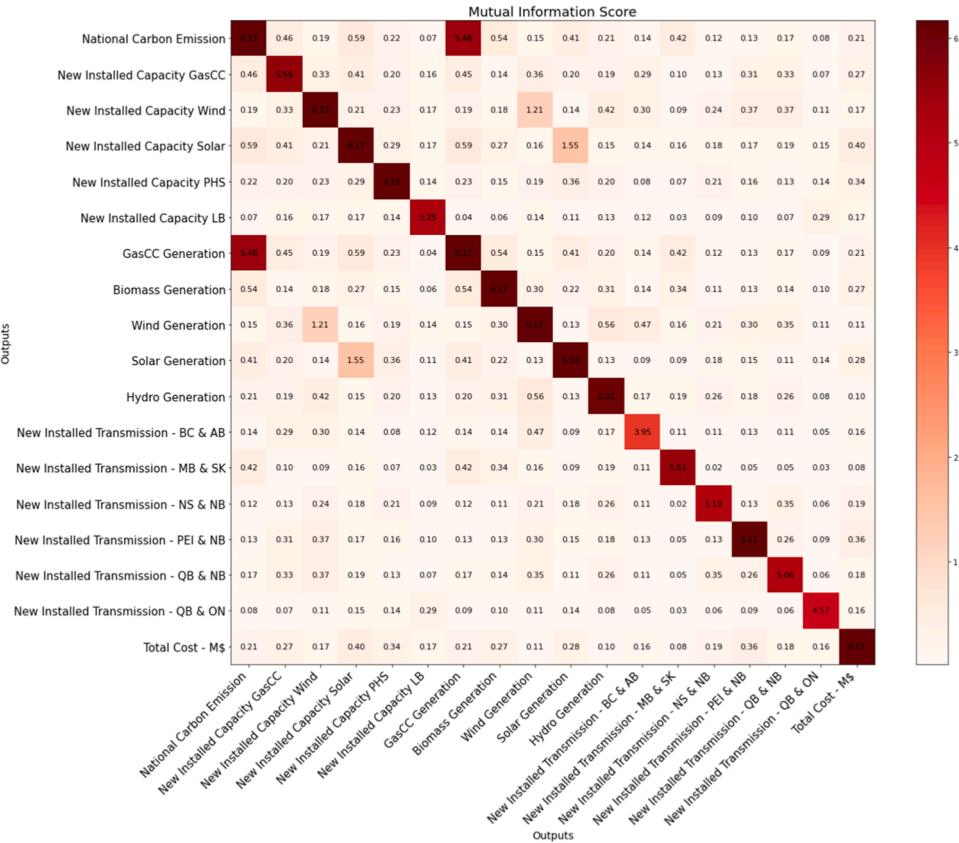


Fig. 3. MI score values between output variables of the Canadian power system in 2020.

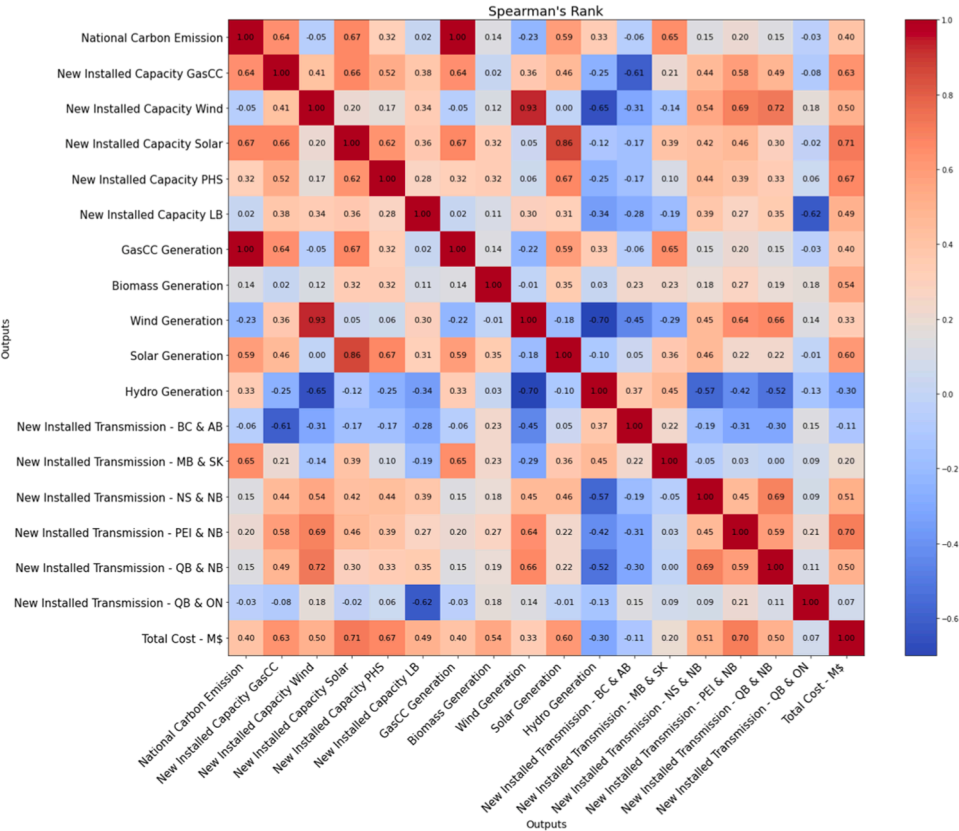


Fig. 4. Spearman's rank values between output variables of the Canadian power system in 2020.

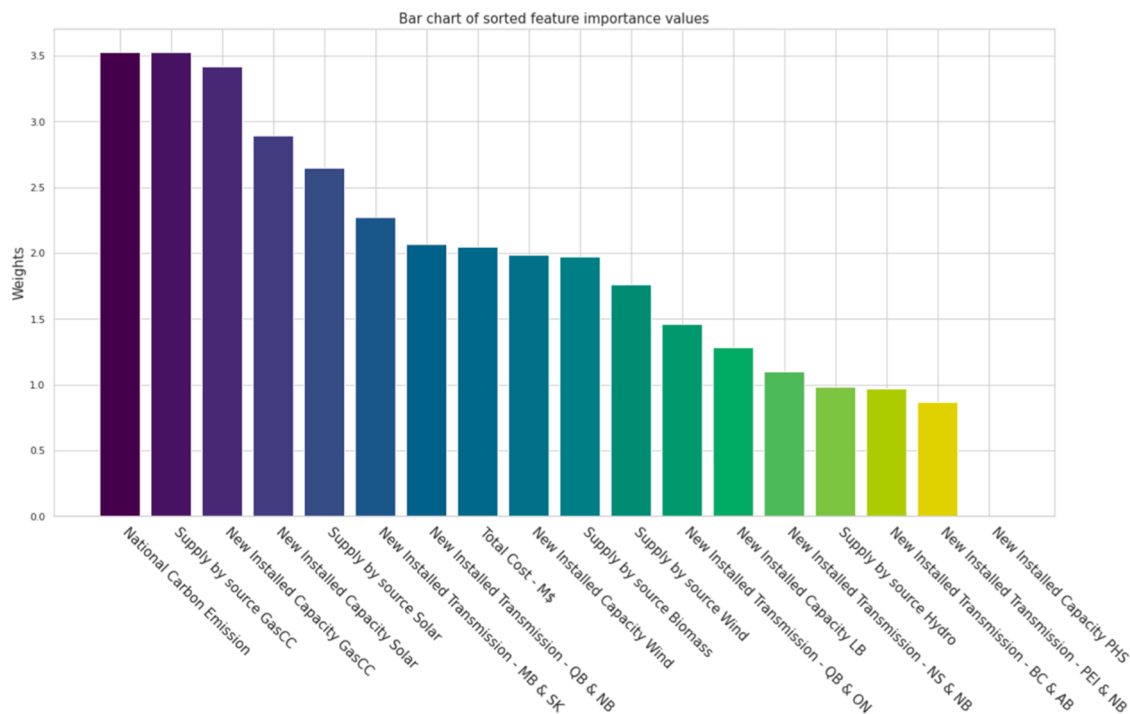


Fig. 5. Importance Weight of output variables in clustering.

energy mix. This is because wind capacities are studied to have a pivotal role in the decarbonization pathways of the Canadian power system.

The spread distribution of solar and wind generation and new installed capacity highlight both the potential and the challenges of renewable energy in the Canadian power system. A high central tendency for both wind and solar generation implies a significant contribution from these zero-emission sources, which is a positive sign for decarbonization efforts.

The costs for wind and solar comprise capital costs, and transmission expansion costs based on the locations' distance to the closest bus. Expanding the wind and solar capacities depends on varying capacity factors and their locations for distance and capacity factor considerations. Therefore, wind and solar capacity expansion tend to be more non-linear (multiple modes in distributions, indicating basic bifurcations and discontinuities) in the output space compared to other variables. Generation is impacted by demand and capacity factors, and comparing new installed capacity and generation plots of wind and solar reveals that they are not equal, underlying the importance of the capacity factor. Additionally, wind generation is less correlated to the installed wind compared to the same couple of variables in solar. This is because building more wind capacity means moving to locations with a lower capacity factor. In the meantime, solar capacity factors for solar are more uniform in different grid locations.

3.4. Hydro power's persistent significance across input assumptions

It appears that hydro generation (min 31 TWh, Max 32.1 TWh, mean 31.9 TWh, std 0.1 TWh) tends to lean towards the higher end ($\sim 42.5\%$ of scenarios), indicating that hydro generation is less sensitive to inputs. Hydroelectric power has an essential role in fulfilling energy demand. It is valuable for its flexibility and cost-effectiveness. Hydropower accounts for 60 % of the total electricity production in Canada, underlining its substantial contribution to the energy system. This contribution is expected to remain significant until 2050 due to the continued operation of existing hydroelectric facilities.

The correlation between hydro and wind output is observed to be negative. As a result of expanded wind capacities, output from hydro,

which serves as a flexible power source, is likely to decrease in order to integrate increased, cheaper electricity generation from wind power.

The total cost of the system is positively influenced by all new build generation types except hydro. The reason is that hydro is not expanding. Additionally, hydro generation ranks among the least deterministic variables in power system scenarios in clustering results, due to its relatively stable, and less variable nature.

3.5. Bifurcation in the gas simple cycle generation

Certain variables display multimodal distributions including gas SC generation (min 12.3 GWh, Max 95.3 GWh, mean 22.3 GWh, std 14.4 GWh) and biomass generation (min 272.4 GWh, Max 2.1 TWh, mean 1.1 TWh, std 0.4 TWh). The multimodal distribution of gas SC generation suggests that there are several possible configurations depending on the parametrization of input variables including the wide range of carbon tax. This multimodal distribution reveals important information:

Peaks with high gas SC generation happen when there is limited availability of low-carbon sources, or there is high demand where gas peaking plants are needed to meet the load. This indicates the need for more renewable generation, storage, or demand response measures to reduce the need for gas SC during these times. As seen when investigating correlations between outputs, such instances occur in scenarios that have a low carbon tax, in which emission-intensive plants, such as gas SC, generate low-cost electricity.

Peaks with low gas SC generation occur when the grid can meet its needs with less carbon-intensive resources. Studying these scenarios provide insights into the conditions that allow for lower gas use, such as high renewable availability or low demand. These instances occur in high carbon tax rates, which limit generation from such plants.

The multimodal distribution generation by gas SC indicates a complex and nuanced role for this power source in the Canadian power system. The generation from gas SC comes from already available assets, while there is no new installed capacity through the whole output space.

3.6. Dominant technologies in the least-cost decarbonized pathways

A few variables showed largely unimodal distributions, albeit with some skewness. These distributions suggest a dominant outcome with variations around it. Variables in this category include new installed capacity of lithium battery (LB) (min 0: ~25 % of scenarios, Max 16.8 GW, mean 2.8 GW, std 2.8 GW), new installed transmission of British Columbia-Alberta, Quebec -Ontario, and Quebec-New Brunswick.

A unimodal distribution of new installed LB storage capacity suggests a prevalent strategy for using LB storage, due to its importance in balancing the intermittency of renewable energy sources. The skewness toward the lower end suggests that based on COPPER's formulation, the required flexibility to integrate large variable wind and solar resources is mainly addressed through other resources, like PHS, and other flexible generation types, such as hydro and gas CC.

3.7. Interprovincial transmissions in the decarbonization pathways

Unimodal distributions are observed for new installed transmission of British Columbia-Alberta (min 0: ~42 %, Max 9 GW, mean 1.5 GW, std 1.8 GW), Quebec-Ontario (min 0: ~34 % of scenarios, Max 4.9 GW, mean 0.9 GW, std 1 GW), Quebec-New Brunswick (min 0: ~25 % of scenarios, Max 1.2 GW, mean 0.2 GW, std 0.2 GW), and Manitoba-Saskatchewan (min 0: ~17 % of scenarios, Max 1 GW, mean 0.2 GW, std 0.1 GW). These variables imply a typical pattern for new transmission installations across these provinces. Transmission capacity expansion is driven by accessing regions with higher Variable Renewable Energy (VRE) potential. This strategy leads the model to build additional wind or solar capacity in optimal locations and then use transmission lines to distribute the generated power to neighboring provinces. The correlation between VRE capacities and transmission expansions in high-potential provinces, such as Alberta, Saskatchewan, Quebec, and Ontario, supports this behavior. However, for some interconnections, this behavior may not be as predictable. If a province experiences a limited load growth, as is the case in New Brunswick, or if two provinces have similar VRE potentials, expansion of transmission capacity is not as robust against possible features.

There appears to be a significant co-occurrence of eastern grid expansions, specifically between Quebec-New Brunswick and Nova Scotia-New Brunswick (min 0: ~25 % of scenarios, Max 515.8 GW, mean 0.9 GW, std 0.9 GW). This suggests that when one expansion occurs, the other is also likely to happen, which makes sense. As enhancing one region's transmission capacity could necessitate upgrades in neighboring areas to accommodate the increased flow of electricity.

The observed minimal correlation among certain output variables, such as the new installed transmission between different provinces and the new installed capacity of LB and PHS, implies their independent behavior. This means that changes in the new transmission capacity of Quebec-Ontario, for instance, do not significantly impact changes in the new transmission capacity of British Columbia-Alberta. Similarly, the decision to install new LB or PHS capacity appears to be independent of the installation of these transmission lines. These low correlations suggest that planning and investment decisions in these specific areas can be made with more flexibility and independence. However, the lack of correlation might also point to potential opportunities for more coordinated planning to maximize system efficiency and resilience.

Eastern grid expansion appears to boost all generation sources except hydro. This again relates to the significant hydro resources that will remain active and will not expand, but other generation capacities tend to expand.

Total cost is positively influenced by all transmission lines except with British Columbia-Alberta transmission. The reason is that British Columbia-Alberta transmission is relatively cheap.

As for new installed transmission lines, Manitoba-Saskatchewan, and Quebec-New Brunswick, exhibit the most influence on shaping different

scenarios, underscoring their ambiguity and pivotal role in power system planning and management. In contrast, Prince Edward Island-New Brunswick (min 0, Max 0.8 GW, mean 0.2 GW, std 0.1 GW) and British Columbia-Alberta demonstrate lesser determinism. This distinction can be attributed to the stronger correlations of the former transmission lines and weaker correlations of the latter with other outputs, as evidenced by the correlation metrics.

Transmission Expansion and VREs. The transmission between western provinces, particularly British Columbia-Alberta and Saskatchewan-Manitoba, appears to affect the pattern of power generation builds. Notably, correlations exist between wind and gas CC for British Columbia-Alberta, and between gas CC and biomass for Saskatchewan-Manitoba.

In the eastern grid, similar dynamics are observed. Transmission build-out, especially Quebec-New Brunswick and Quebec-Prince Edward Island, seems to influence power generation choices, particularly between gas CC and wind.

Transmission build between Quebec-Ontario primarily affects LB build. This is attributed to access to Quebec's hydro resources, potentially reducing the need for extensive energy storage.

Wind supply is strongly correlated with Alberta-British Columbia corridor expansion. Interestingly, the corridor plays a crucial role in providing access to Alberta's high wind potential locations, and British Columbia's highly flexible hydro capacities.

These observations prove the strong correlations of VRE and transmission build between provinces and the reason for these correlations is VRE capacity factors, which force the model to build out in locations that have more potential in general. Based on that, the model decides to build more wind/solar in one location with more potential and transfer it to neighboring provinces. Canada's power system decarbonization is dominated by being able to access high VRE regions of the country, and this distributes that generation across the country with transmission and back it up with hydro. The VRE capacity factor limitation for transmission will impact gas CC generation choices as the non-renewable energy source. Similarly, a correlation between gas CC and solar power generation is observed, suggesting these generation types often be used in combination within the power system.

3.8. Uncertainty in the future trajectory of storage assets

A multimodal distribution with a left extremal tendency for new installed capacity of pump hydro storage (PHS) (min 0, Max 12.5 GW, mean 1.6 GW, std 1.4 GW) reflects various potential operational scenarios or strategies. The left extremal tendency (~45 % of scenarios) suggests that there are cases where significantly less new PHS capacity is installed than average, indicating scenarios where other types of storage or generation options may be more optimal. This shows the importance of considering a diverse set of options in the planning and optimization of energy systems. It is also indicating that there is still a lot of uncertainty in the future of storage in power systems.

New installed LB and PHS show positive correlations with gas CC, solar, and wind. This indicates that these storage technologies are often built in conjunction with these generation sources to balance their output and ensure grid reliability. Gas CC is another flexible capacity to support the requirements to integrate larger wind and solar capacities at lower carbon tax rates.

The installed capacity of LB carries more weight than PHS in shaping different scenarios, which doesn't impact the selected pathways and outputs of the COPPER model. This is due to the cost uncertainty associated with LB.

3.9. Exclusion of nuclear and diesel generation in the future grid development

Nuclear (min 5.9 TWh, Max 30.4 TWh, mean 8 TWh, std 3.1 TWh), in ~85 % of scenarios and diesel, in ~72 % of scenarios, are not built or

generated in most future grids. Furthermore, generation and expansion outputs are in a much relatively smaller range in the rare cases they are present. These variables exhibited largely non-stochastic or deterministic behavior, suggesting these outputs are less influenced by variations in input variables. Nuclear is not sensitive, as its capital cost is much higher than other resources, thus, the model does not build it in many cases. For nuclear energy, which plays a significant role in decarbonization due to its zero-carbon emissions, this predictability aid in long-term planning and grid stability. Diesel is an emission intense resource. It is most likely that the model doesn't build out any diesel in most of the carbon taxation inputs.

3.10. Drivers for carbon emissions

Spread distribution of national carbon emissions reflects the inherent variability of this output variables. This is because of the wide range of carbon tax policy affecting the emission.

Carbon emissions have a strong association with gas CC. After gas CC, solar, biomass, and installed transmission of Manitoba-Saskatchewan affect national emissions the most. This highlights these factors as key contributors to the system's carbon footprint. While gas SC mainly plays as a peaking capacity in higher carbon tax rates, gas CC remains the main contributor to emissions in lack of other emitting sources which retired due to high emission rates and carbon taxation. These correlations help in identifying the most effective strategies for emissions reduction. If certain renewables, solar and biomass, or transmission enhancements of Manitoba-Saskatchewan are more strongly correlated with emission reductions than others, these might be prioritized in future system development or policy efforts.

3.11. Factors shaping divergence in the energy transition pathways

The carbon emission levels, generation and the new installed capacity of gas CC, with the maximum feature importance weight (~ 3.5), and solar power systems, (feature importance weight 2.8) have the most influence on creating diverse scenarios within the Canadian power system in 2050. These factors play a pivotal role in shaping the various pathways and outcomes observed in the different scenarios.

Notably, all these output variables are among the most correlated outputs based on the correlation metrics values presented in heatmaps. The reason is that stronger correlations indicate a greater number of variables influencing a specific output, consequently affecting more output variables and shaping diverse scenarios. This introduces more variations and uncertainties. Therefore, these variables are more influential in the planning of different energy pathways and demand careful consideration.

4. Limitations

Like all modeling methods, our approach also has certain limitations that originate from two main sources; the COPPER model and the ML surrogate, other than the limitations mentioned for these two models in previous studies (Arjmand and McPherson, 2022a; Jahangiri et al., 2023b).

Firstly, the COPPER model employs a fixed capacity factor and does not account for capacity transfers, i.e., the process of constructing a power plant in one location and contracting to sell its capacity to another jurisdiction. The model's focus is primarily on energy transfer, resulting in capacity for transmission construction that is majorly driven by the availability of VRE potentials. COPPER also simplifies assumptions by treating transmission and most generation technologies as continuous variables rather than binary integers. A more realistic approach would involve modeling contemplated projects with known capacities, potentially through regional analysis to reduce computational burdens. Additionally, COPPER does not consider landscape characteristics that could affect wind and solar development, may

overstate development potential in some regions, and has limited balancing areas and intraprovincial transmission expansion (Arjmand and McPherson, 2022b).

Secondly, the limitations related to the ML surrogate pertain to its dependence on the current COPPER version. As the COPPER model evolves and improves over time, re-training of the surrogate model becomes necessary to ensure that the changes are accurately captured and integrated into our method. The reliability of the surrogate model depends on the quality and comprehensiveness of the training data, with potential inaccuracies when extrapolating beyond the training data. The ML model also exhibit less sensitivity to parameters with small nominal values, potentially affecting predictive accuracy for certain provinces. Finally, while the surrogate model provides predictive power, understanding the specific reasons behind its predictions may be challenging, impacting interpretability (Jahangiri et al., 2024).

5. Discussion

This paper presents significant advancements in the field of large-scale power system planning, notably through the application of ML methods to overcome computational limitations of conventional modeling approaches for uncertainty analysis. The existing literature primarily focuses on utilizing ML for predictions and optimization in various fields. While there are studies that examine uncertainty analysis using ML to estimate prediction errors, there is a noticeable lack of research on standardized uncertainty analysis by exploring the output space of large-scale power systems. The focus of uncertainty analysis is not on prediction, but on exploring and characterizing the output space by comprehensively understanding the behavior of the system through analyzing results probabilistically, considering input uncertainty, and exploring correlations among outputs. This probabilistic analysis provides insights into system behavior and its implications and enables the exploration of key output variability. Additionally, the exploration of output correlations uncovers relationships and dependencies among variables, enriching the understanding of complex system interactions. This holistic approach enhances decision-making capabilities and facilitates predictions under various scenarios, contributing to a comprehensive characterization of model behavior.

Exploring output interdependencies lead to more effective power system planning. For example, understanding the least correlated variables help identify areas of independent planning or where there are opportunities for improved coordination. On the other hand, knowledge of the most correlated variables aids in predicting the system-wide impacts of changes in a specific area and in making more integrated, sustainable planning decisions. In terms of decarbonization, recognizing the variables that correlate strongly with national carbon emissions guide strategies to reduce these emissions.

Lastly, the presence of some non-intuitive correlations among the output variables can be attributed to the underlying demand growth, indicating that demand-side factors influence these relationships. The observed correlations primarily stem from the stochastic demand as an input, rather than indicating direct relationships. While further investigation into this matter is beyond the scope of this discussion.

6. Conclusion

This study emphasizes the importance of uncertainty analysis in energy systems, particularly in large-scale power systems, and highlights the limitations of traditional methods. The research identifies gaps in applying ML to analyze complex power systems and calls for exploratory ML approaches to gain valuable insights. The use of neural networks as surrogate models is proposed to enhance uncertainty analysis and inform policy planning. The paper aims to standardize uncertainty analysis in power systems by extending ML methods for input and output space exploration to address research gaps and explore output space correlations, ultimately contributing to effective decision-

making in complex socio-technical systems. This study brings a new perspective to power system modeling, offering valuable insights into system behavior, causative factors, and distinct decarbonization pathways. The proposed approach forms a robust foundation for strategic planning and policy formulation to achieve decarbonization targets. The most important takeaways of this study are:

- Further research investment is crucial in incorporating uncertainties associated with capacity factors, as they have shown a significant impact on power system outcomes.
- Gas CC power (generation mean 4.8 TWh, installed capacity mean 4.8 GW) emerged as a more flexible option than gas SC (generation mean 22.3 GWh 0 installed capacity), exhibiting a lower impact from carbon taxation due to its lower emission factor. While gas CC is less impacted by carbon tax, it is utilized in the model to support large wind and solar integration even at very high tax rates. While the emissions are low in this technology, deep decarbonization of the power systems dictates other policies in addition to carbon taxation.
- Hydroelectric power, a significant contributor to Canada’s energy mix, showed less sensitivity to input changes and remained relatively stable (in ~42.5 % of scenarios). This technology will provide system-wide flexibility for integrating large VRE expansions and remains one of the main contributors to support the decarbonization of the power system. However, its expansion is less impacted by the selected input parameters and their range. Hydroelectric remains important but must be maintained and developed in a new way that respects the very legitimate concerns that currently face it i.e., disclosure of Site C dam.
- Transmission expansion patterns reflect the influence of VRE capacity factors and their difference in different locations. Correlations, like that between Alberta’s wind potential and the Alberta-British Columbia line, reinforce this point. The VRE capacity factor limitations also had ripple effects on non-renewable source choices, favoring combinations like gas CC with wind and solar. This insight also emphasizes the importance of transmission expansion and power systems integration in the decarbonization pathways.
- Nuclear and diesel powers respectively in ~ 85 % and ~ 72 % of scenarios, owing to their high cost and emission respectively, were less favored, emerging as almost non-stochastic variables. With no cap on emissions, these two technologies are not selected to provide firm capacities even in higher wind and solar penetration rates. As such, nuclear is not selected over gas CC even in the lower capital cost inputs. And diesel capacities are also not expanded and used even in lower carbon tax rates. Therefore, policymakers should not bet on nuclear and diesel, and power systems’ deep decarbonization requires implementing extensive regulations along with carbon taxations.
- Carbon emissions, gas CC and solar powers are key differentiators in future power system scenarios. Solar held a more significant role than wind in these differentiations. For storage, the installed capacity of LB was more influential than PHS, while in transmission lines, Manitoba-Saskatchewan and Quebec-New Brunswick had the most and Prince Edward Island-New Brunswick and had British Columbia-Alberta the least influence. These insights provide valuable guidance for policymakers, highlighting the key areas that require focused attention for effective planning and decision-making.
- Further research investment is needed to better understand and forecast demand, as it drives pathways, contributes to non-intuitive correlations, and remains uncertain.

The findings developed using ML and numerical analysis have potential implications that can be further explored in future research. This includes incorporating uncertainties associated with wind and solar capacity factors to refine power system modelling. Furthermore, a provincial-level application of the methodologies developed in this

study could lead to more tailored insights for effective provincial policy planning. In considering future research directions, it’s essential to acknowledge additional factors that may influence power system modeling accuracy and policy planning outcomes. For instance, variations in the quality of wind and solar resources, distance to load centers, landscape characteristics, and interprovincial transmission costs can significantly impact grid decarbonization strategies. While current methodologies like COPPER accurately quantify resource quality and transmission costs, there’s potential to explore the effects of land availability and interprovincial transmission capital costs on decarbonization efforts further. Addressing these factors presents an opportunity to refine modeling techniques and develop more tailored policy interventions. However, challenges persist in validating data accuracy, managing computational demands, and ensuring model scalability.

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CRediT authorship contribution statement

Mohammad Miri: Writing – review & editing, Conceptualization. **Zahra Jahangiri:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kwang Moo Yi:** Writing – review & editing, Supervision, Project administration. **Madeleine McPherson:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

We made the data set publicly available

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Abbreviations

Symbol	Definition
AB	Alberta
BC	British Columbia
SK	Saskatchewan
MB	Manitoba
ON	Ontario
QC	Quebec
NB	New Brunswick
PEI	Prince Edward Island
NS	Nova Scotia

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